SkillShift

An AI career news portal that traces tech news to real TPM hiring requirements

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> ***The idea in one line:*** *news tells you what changed, job postings tell you what is hired for. SkillShift connects the two, and lets you click through to the evidence behind every insight.*

1. Problem

Tech job seekers read a constant stream of news but cannot tell which shifts matter for hiring. News portals summarize what happened without saying what it means for a career. Resume tools match keywords to postings without explaining why a skill matters now. Generic chatbot advice ("learn AI evaluation") comes with no evidence the user can check.

2. Target User

Graduate students and early-career professionals preparing for Technical Program Manager (TPM) internships and full-time roles, who must decide what to learn, what to highlight on a resume, and what to prepare for interviews, with limited time.

3. Solution

SkillShift reads curated tech news, groups related articles into shared themes, and connects each theme to the skills real TPM postings ask for. The output is a sourced career briefing where every insight traces back to the articles and postings behind it. **News is the starting point; job postings are the evidence that grounds it.**

4. Key Features

Built in this prototype

- **News-to-Skill Traceability:** click a skill to see related articles, postings that require it, and the AI interpretation, shown separately
- **Cross-article themes:** 2 to 3 shared themes from multiple sources, not article-by-article summaries
- **Sourced briefing:** What changed / Why it matters for TPMs / What to do next, every claim linked to a source
- **Human review:** the user approves or excludes AI findings before they appear in the briefing
- **Clear labeling:** verified evidence vs AI interpretation; observed signals vs established trends

Future: more roles (PM, SWE, Data Science), live feeds, audio briefing, personal skill-gap recommendations

5. AI / Technical Approach

A role-separated pipeline on the Claude API, built with Next.js and TypeScript. Each agent has a defined input, output, and validation step.

- **Scout Agent:** extracts TPM-relevant topics and clusters similar articles into themes
- **Signal Agent:** compares themes against skills in TPM postings and keeps only evidence-backed links
- **Editor Agent:** writes the briefing, keeping verified evidence and AI interpretation visibly separate
- **Human Review:** approve or reject items before publication

Flow: curated articles + TPM postings → Scout (themes) → Signal (skill links + evidence) → Editor (draft briefing) → Human Review → published briefing.

Data: 5 to 7 recent tech news articles and 3 to 5 TPM job postings, curated manually. Outputs are schema-validated and precomputed for demo reliability.

6. Demo Flow

- Persona: a CMU student preparing for TPM recruiting
- Today's briefing: 2 to 3 themes detected across the news
- Click a skill and trace it to source articles and postings
- Review queue: approve or exclude a finding, briefing updates
- Pipeline view: how Scout, Signal, and Editor pass data

7. Limitations

- Small hand-curated dataset: findings are observed signals, not statistically validated market trends
- LLMs can misread or over-connect; human review and source links reduce this but do not eliminate it
- Limited to TPM roles and English-language sources

8. Future Direction

As a semester team project, **each member owns a role vertical** (PM, SWE, Data Science), so the product grows with the team's own career interests. Next: larger and live data, trend validation over time, personalized learning paths, and testing with CMU students during recruiting season. **Business model:** licensing to university career centers and bootcamps, plus a free tier for individual job seekers.